Rowing Metrics

User Guide — Screens & Options

Cross-platform rowing session tracker for Android and iOS.
This document describes every screen, metric, and configuration option in the app.

1. Overview

Rowing Metrics is a rowing session tracker that uses your phone's accelerometer to detect stroke rate (SPM) and GPS to measure speed and distance. The app is built with Kotlin Multiplatform and shares the same interface on Android and iOS.

Key capabilities:

- Live stroke rate, speed, average speed, distance, and elapsed time during a session
- Configurable stroke detection sensitivity and GPS speed smoothing
- Automatic saving of completed sessions to a local history
- Export activity history to CSV (Excel-compatible) on Android
- Multi-language UI: English, Catalan, Spanish, and French

Navigation

The app has four screens arranged in a horizontal pager. Swipe left or right to move between them:

#	Screen	Purpose
1	Main metrics	Live session display — Start/Stop rowing
2	Configuration	Stroke sensitivity, speed smoothing, language
3	Activities	History of completed sessions
4	About	App name, version, and contact information

2. Main Metrics Screen (Screen 1)

This is the primary on-water display. It shows live metrics while a session is running, and the final values after you press Stop. The layout adapts automatically to portrait and landscape orientation.

Metrics displayed

Field	Unit / format	Description
Stroke rate	SPM (integer)	Strokes per minute from accelerometer detection. Largest value on screen.
Speed	km/h (1 decimal)	Current speed from filtered GPS. Smoothed using your Speed smoothing setting.
Average speed	km/h (1 decimal)	Total distance ÷ elapsed time for the current session.
Distance	meters (integer)	Cumulative GPS distance since Start. Only counts movement "e 4 m per segment.
Time	MM:SS or HH:MM:SS	Elapsed session time from Start to Stop. Shows hours once "e 1 hour.
Clock	HH:MM	Device local time (updates every second). Shown beside Time.

Start / Stop button

A large green Start button begins a new session. While running, it turns red and shows Stop.

- Start: begins accelerometer stroke detection, GPS tracking, and the session timer
- Stop: ends the session, saves a summary to Activities history, and resets live metrics
- On first Start, Android requests location permission (Fine and Coarse) if not already granted

Stroke detection warning

While a session is running, if no stroke has been detected recently, the stroke rate band background blinks between light red and black. This alerts you that the app is not registering strokes — you may need to adjust phone placement or increase detection sensitivity in Configuration.

Portrait layout

Top to bottom: Stroke rate (largest band), then Speed, Average speed, Distance, Time + Clock side by side, and Start/Stop at the bottom.

Landscape layout

Left column: Stroke rate and Start/Stop button. Right column: Speed and Average speed on one row, then Distance, then Time + Clock. Text is scaled up for readability on wider screens.

Android-specific behavior

- Screen stays on while the app is in the foreground (FLAG_KEEP_SCREEN_ON)
- App can appear over the lock screen and turn the screen on — useful while rowing
- Requires device gyroscope/accelerometer for stroke detection

3. Configuration Screen (Screen 2)

Title: "Rowing Metrics Configuration". Settings are stored immediately when changed and apply to the next session (stroke sensitivity changes mid-session take effect after Stop).

Stroke profile — Detection sensitivity

A 5-step horizontal slider controls how aggressively the accelerometer accepts stroke events. Left = strictest; right = most permissive. Only three positions show text labels; two intermediate steps are unlabeled tick marks.

Position	Label	Behavior
1 (left)	Low	Strictest — fewest false strokes; soft strokes may be missed
2	(tick only)	Between Low and Medium — constructor defaults
3 (center)	Medium	Default — balanced blend between permissive and strict
4	(tick only)	Between Medium and High — former "High" preset
5 (right)	High	Most permissive — easiest stroke acceptance; higher false-positive risk

Move the slider toward High (right) if strokes are not being detected. Move toward Low (left) in rough water or when handling the phone causes spurious counts. Medium works well for most rowers.

Speed smoothing

Slider range: 1 to 8 (default 4). Controls how many recent filtered GPS speed samples are averaged for the live Speed readout.

- 1 = most responsive — uses only the latest filtered sample
- 8 = smoothest — averages the last 8 filtered samples

Higher values reduce GPS jitter but add lag to speed changes. The setting does not affect Average speed (which is always distance ÷ time).

Language

Dropdown to select the UI language. Available options:

- English (default)
- Catalan
- Spanish
- French

All labels, buttons, dialog text, and CSV export column headers use the selected language.

4. Activities Screen (Screen 3)

Title: "Activities". Shows a scrollable table of all completed rowing sessions saved on the device. Sessions are created automatically when you press Stop on the main screen.

Table columns

Column	Content
Date	Date the session ended
Hour	Start time of the session (derived from end time " duration)
Time	Session duration (MM:SS or HH:MM:SS)
Stroke rate	Average SPM for the session
Avg. speed	Average speed in km/h
Distance	Total distance in meters
Delete	Tap to remove that single session

Empty state

If no sessions exist yet, the table shows: "No activities yet. Press Start on the first screen, then Stop to record one."

Export activities (Android)

When at least one activity exists, a download icon appears in the top-right corner.

- Tap the icon !’ confirm "Export to Excel" dialog
- Choose a save location; a CSV file is written with all sessions
- Open the CSV in Excel, Google Sheets, or any spreadsheet app

Exported columns: Date, Hour, Time, Stroke rate, Avg. speed (km/h), Distance.

Delete all

A "Delete all" link appears next to the export icon when activities exist. Tapping it shows a confirmation dialog. This permanently removes every row — it cannot be undone.

Delete individual session

Tap "Delete" on any row to remove that session immediately (no confirmation dialog).

5. About Screen (Screen 4)

The fourth screen (swipe left from Activities) shows a card with:

- App title: "Rowing Metrics"
- Version number (e.g. v2 — matches the installed app version)
- Contact email: gardenyes@gmail.com

6. How Metrics Are Calculated

This section explains what the numbers mean behind the UI.

Stroke rate (live SPM)

- Accelerometer samples at ~40 Hz detect stroke peaks using adaptive thresholds
- SPM is recalculated approximately every 2 seconds from recent stroke intervals
- Requires at least 8 detected strokes before a stable SPM is shown
- SPM returns to 0 if no stroke is detected for ~3 seconds
- Session average SPM = mean of SPM readings sampled every ~2 s while running (not total strokes ÷ time)

Speed (current)

- Raw GPS speed (m/s) passes through a Kalman filter, then a FIFO buffer (max 8 samples)
- The Speed smoothing setting (1–8) controls how many of the latest filtered samples are averaged
- Result is converted to km/h (× 3.6) for display
- GPS fixes with accuracy > 50 m are rejected; speed shows 0 until GPS is ready

Average speed

Computed as total session distance (km) ÷ elapsed time (hours). Not an average of instantaneous speed readings.

Distance

- Accumulated from GPS position changes using haversine distance
- Only segments of "e 4 meters are counted (filters GPS noise when stationary)
- Large GPS jumps (> 200 m with poor accuracy) are ignored

Orientation changes

If the phone orientation changes during a session, stroke detection pauses for ~1.2 seconds to stabilize, then resumes automatically.

7. Requirements & Platform Notes

Permissions (Android)

- Location (Fine or Coarse): required for speed and distance during a session
- Motion sensors: required for stroke detection (no separate permission on Android)

Platform differences

Feature	Android	iOS
Stroke detection	Full support	Stubbed — pending Core Motion bindings
GPS speed/distance	Full support	Stubbed — pending Core Location bindings
Activities history	Full support	Full support (SQLDelight)
Configuration / language	Full support	Full support
CSV export	Full support	Not yet implemented
Lock screen display	Yes	Platform default

Hardware

- GPS-capable device for speed and distance
- Accelerometer/gyroscope for stroke rate detection

8. Quick Reference

Screen swipe order

Main metrics !’ Configuration !’ Activities !’ About !’ (wraps) Main metrics

Default settings

Setting	Default
Detection sensitivity	Medium (center position)
Speed smoothing	4
Language	English

Typical workflow

1. Open the app (grant location permission when prompted).
2. Optionally swipe to Configuration and adjust sensitivity or language.
3. On the main screen, press Start when you begin rowing.
4. Monitor stroke rate, speed, and distance live.
5. Press Stop when finished — the session is saved to Activities.
6. Swipe to Activities to review, export, or delete past sessions.